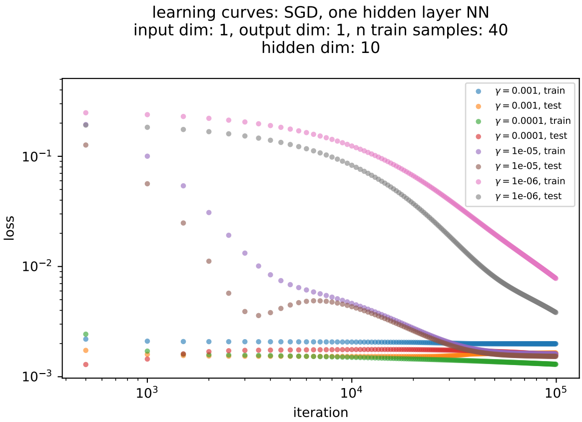
Machine learning Tek 5: simplicity bias of neural networks



Data

Polynomial regression and overfitting

Neural network regression

We will perform regression on this data :

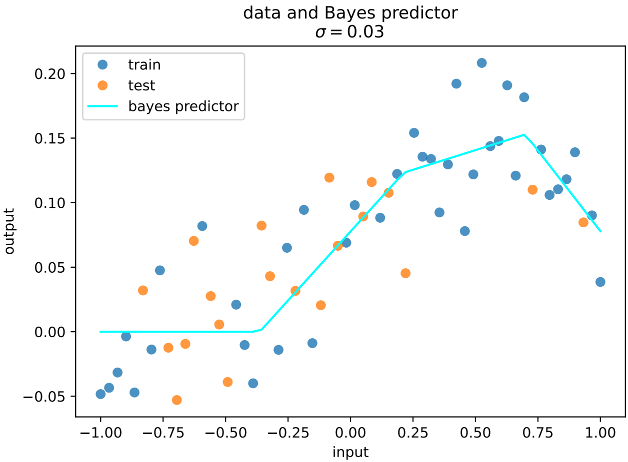


Figure – data

We have seen that polynomial regression without regularization can **overfit** :

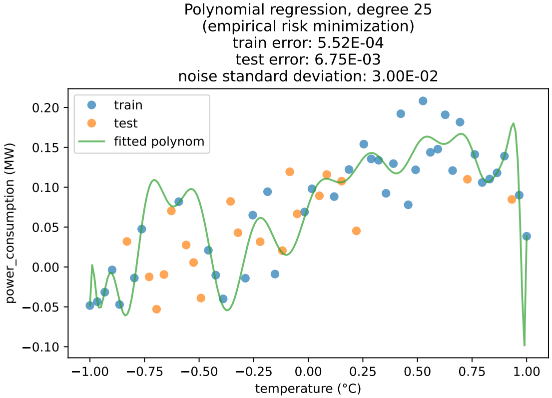


Figure – Overfitting of a neural network with 26 parameters

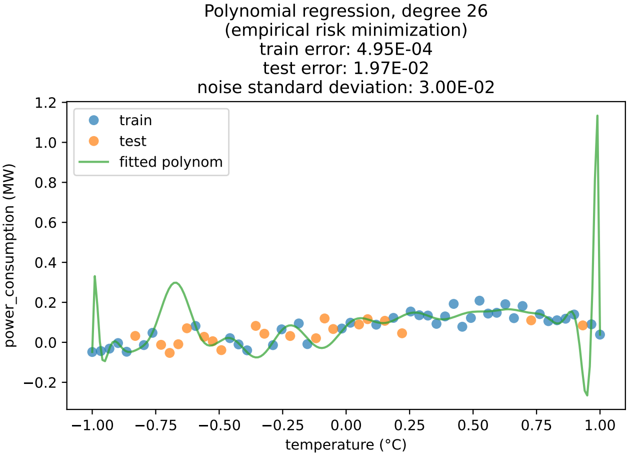


Figure – Overfitting of a neural network with 27 parameters

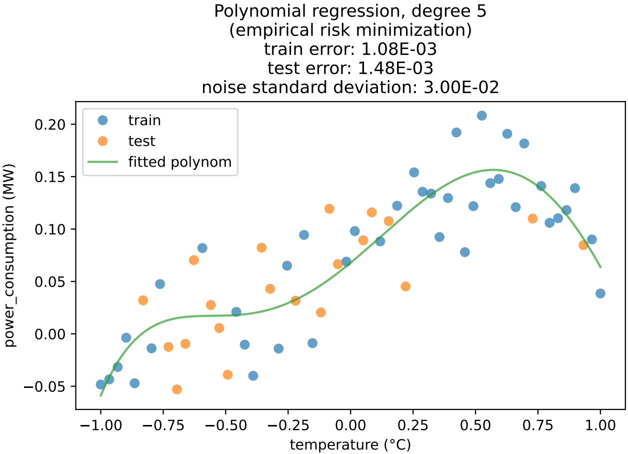


Figure – With only 6 parameters, polynomial regression does not overfit.

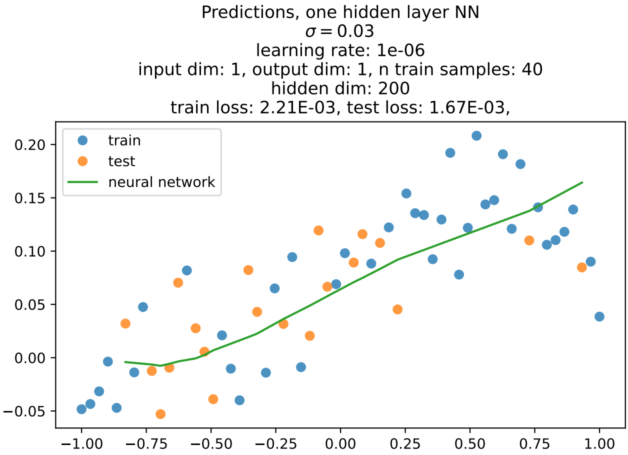


Figure – Even with 400 parameters, this neural network trained with SGD did not overfit !

Exercice 1 : Implement neural network with 1 hidden layer, with a variable number of hidden neurons, and study the amount of overfitting, also as a function of the learning rate.

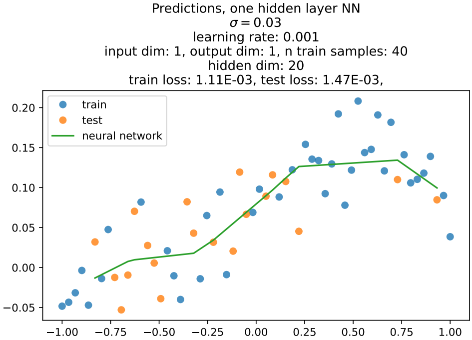
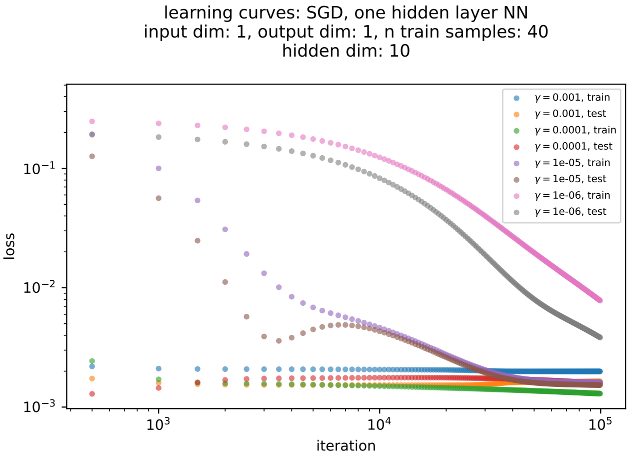
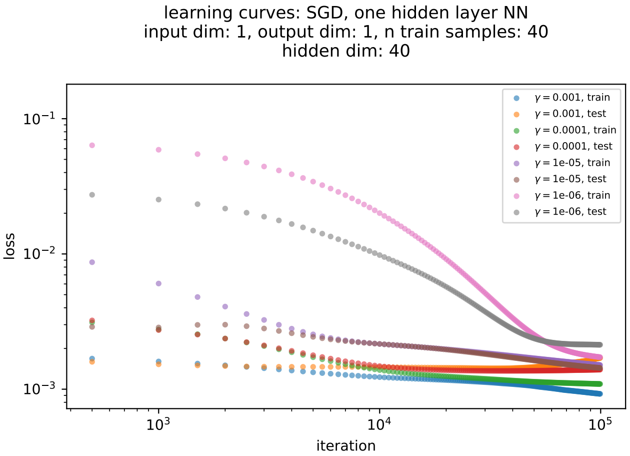


Figure – A neural network with 40 parameters

Learning curves



Learning curves



Learning curves